

A TRAFFIC ACCIDENT CAUSATION ANALYSIS METHOD BASED ON AHP- APRIORI ON INDIAN CONTEXT

PROJECT REPORT

Submitted by

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in

INDUSTRIAL ENGINEERING AND MANAGEMENT



**DEPARTMENT OF MECHANICAL ENGINEERING
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MAY 2026

DECLARATION

I undersigned hereby declare that the Project report “**A TRAFFIC ACCIDENT CAUSATION ANALYSIS METHOD BASED ON AHP- APRIORI ON INDIAN CONTEXT**” submitted for partial fulfilment of the requirements for the award of degree of Master of Technology of the APJ Abdul Kalam Technological University, Kerala is a Bonafide work done by me under the supervision of **Prof. Devanand C N**. This submission represents my ideas in my own words and where ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources. I also declare that I have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in submission. I understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

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CERTIFICATE

This is to certify that the Project Report entitled “**A TRAFFIC ACCIDENT CAUSATION ANALYSIS METHOD BASED ON AHP- APRIORI ON INDIAN CONTEXT**” submitted by **ABHIJITH M** to the APJ Abdul Kalam Technological University in partial fulfilment of the requirements for the award of the degree of Master of Technology in Industrial Engineering And Management is a Bonafide record of the project work carried out by him under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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ABSTRACT

Road traffic accidents remain a leading cause of death and serious injury in India, with nearly 1.19 million lives lost globally each year. Despite the availability of large accident databases, most existing studies either rely on subjective expert judgment or analyse individual factors in isolation, failing to capture the multi-factor nature of real-world accidents. This study proposes a data-driven hybrid framework combining the Analytic Hierarchy Process (AHP) and the Apriori algorithm to identify and rank the key factors contributing to road accident severity using 15,000 Indian road accident records spanning 2018 to 2023. The central innovation of this work is the replacement of traditional subjective Saaty-scale expert judgment in AHP with statistically computed Cramér's V values, making the factor ranking process fully objective and reproducible. The AHP analysis identified Vehicle Type Involved as the most influential factor with a global weight of 0.3275, followed by Lighting Conditions at 0.1825. The Apriori algorithm subsequently revealed that dark lighting combined with dry road surfaces produces the highest fatal accident rate at 34.3%, and that pedestrians in stormy weather face the greatest risk at 36.8%. The findings demonstrate that multi-factor analysis uncovers critical accident patterns invisible to single-variable approaches, and the proposed framework provides a reliable, evidence-based tool for road safety planning and policy intervention in the Indian context.

Keywords: Road accident analysis, AHP, Apriori, Cramér's V, Accident severity, Data mining, India

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CHAPTER 1

1.1 INTRODUCTION

Road traffic accidents remain one of India's most critical public health and infrastructure challenges. According to the World Health Organization (WHO, 2023), India accounts for a disproportionate share of global road fatalities relative to its vehicle population. The Ministry of Road Transport and Highways (MoRTH) collect vast accident data annually, yet this data is largely used for statistical reporting rather than predictive or causal analysis.

The need for a scientific, data-driven approach to accident causation analysis is clear. Traditional studies focus on single-variable analysis or rely on expert judgment — approaches that fail to capture the multi-factor nature of real-world accidents. A single accident is rarely caused by one factor in isolation; rather, it is the simultaneous interaction of driver behaviour, vehicle type, road condition, and environmental factors that leads to a severe outcome.

This project addresses this gap by applying a hybrid AHP–Apriori methodology to Indian road accident records. The Analytic Hierarchy Process (AHP) is used to rank accident-causing factors by statistical importance, while the Apriori algorithm mines association rules that link combinations of these factors to accident severity outcomes (Fatal / Serious / Minor).

The primary innovation of this study is the replacement of subjective expert judgment in the AHP pairwise comparison matrix with objectively calculated Cramér's V values – a measure of the statistical association between each factor and accident severity. This makes the analysis fully reproducible and grounded in the data itself.

1.2 PROBLEM STATEMENT

Existing Indian Road accident databases collect vast quantities of data spanning driver demographics, vehicle types, road conditions, and environmental factors. However, this data is predominantly used for frequency-based statistical reporting. The hidden multi-factor correlations that collectively drive fatal and serious accidents, such as the compounding effects of poor lighting, high-speed zones, and heavy-vehicle involvement remain unexplored.

Furthermore, existing analysis frameworks either rely on subjective expert judgment (introducing personal bias) or apply data mining without prior factor weighting (generating

noisy, low-quality rules). A scientific, reproducible, data-driven multi-factor analysis method that is specifically calibrated to Indian road conditions is urgently needed.

1.3 OBJECTIVE

The main objective of this study is to analyse Indian road accident data from 2018 to 2023 using a hybrid AHP–Apriori method on a dataset containing 15,000 accident records. Instead of using expert opinions in the AHP method, this study uses Cramér’s V statistical values to make the analysis more data-driven and objective. The study aims to identify the major factors that contribute to road accidents and rank them based on their influence. It also uses the Apriori algorithm to discover relationships between different accident conditions and accident severity. Based on the findings, the study provides practical recommendations that can help Indian road safety authorities improve traffic safety and reduce accidents.

1.4 SIGNIFICANCE OF THE STUDY

This study uses a dataset containing 15,000 Indian road accident records with 22 variables have been collected. All analyses, findings, and recommendations are developed specifically for the Indian road and traffic environment. The study mainly focuses on categorical accident factors, while continuous variables such as age and speed are converted into meaningful groups or categories for easier analysis. Instead of examining individual factors separately, the research aims to identify relationships between multiple accident factors together to understand how different conditions combine and influence accident severity.

CHAPTER 2

LITERATURE REVIEW

The foundational work for this study is the hybrid AHP–Apriori framework proposed by Xi et al. (2016). Their study applied the Analytic Hierarchy Process to rank accident-causing factors and subsequently used the Apriori algorithm to discover association rules between accident conditions and severity outcomes, using Chinese road accident data. The study demonstrated that driver, road, environment, and vehicle are the four major accident categories, and that environmental conditions show strong associations with accident severity. However, a critical limitation of this work is that AHP rankings were derived from subjective expert judgment using Saaty's 1–9 scale, introducing personal bias into the factor weighting process. Additionally, the framework was not validated against Indian road conditions or the complexities of mixed-traffic behaviour prevalent in developing countries.

Building on the standalone application of Apriori, several studies have explored association rule mining for road accident analysis without integrating a factor prioritization step. One such study, published in a road safety data mining journal, applied the Apriori algorithm directly to accident datasets and successfully identified patterns such as overspeeding, mechanical failure, and unsafe overtaking as key accident contributors, using support and confidence metrics to rank rule significance. However, the absence of any pre-mining factor weighting or dimensionality reduction meant that all variables were treated as equally important, generating a large volume of rules — many of which carried little practical significance.

A more statistically rigorous approach was taken in a study on fatal accident analysis using data mining techniques, which combined statistical analysis, clustering, and Apriori association mining to examine the relationships between weather conditions, road surface characteristics, collision type, and fatal accident rates. The study found that human factors such as drunk driving and collision type exerted a stronger influence on fatal outcomes than environmental factors alone. Despite its depth, this study lacked a unified hierarchical framework integrating human, vehicle, road, and environmental factors together, and did not employ any multi-criteria prioritization technique before rule mining.

Addressing computational efficiency, another study explored the combination of K-Mode clustering with the Apriori algorithm to classify accident records into meaningful groups before extracting association rules. Published in a data mining and transportation research journal, this work showed that pre-clustering the data significantly improved Apriori efficiency and enhanced rule quality by reducing noise in the input dataset. However, the study focused primarily on clustering accuracy and rule extraction performance, without establishing the relative importance of accident causation factors through any decision-making methodology such as AHP.

Finally, a study focused specifically on rainy-weather road crashes applied Apriori association rule mining to examine the influence of road geometry, illumination conditions, and driver age on crash severity during precipitation. The findings revealed that poor illumination and curved roadways significantly increased crash severity in rainy conditions, and that young drivers showed a higher probability of run-off-road crashes under reduced visibility. While the study provided valuable weather-specific insights, its scope was restricted to a single weather condition, limiting the generalizability of its findings across diverse environmental and road scenarios.

Taken together, the reviewed literature establishes both the individual effectiveness of AHP and Apriori and the value of their hybrid combination. However, no prior study has applied a fully data-driven AHP framework - one in which pairwise comparison values are derived from statistical measures rather than expert opinion - to Indian road accident data. This gap in the literature forms the direct motivation for the present study.

CHAPTER 3

THEORETICAL FRAMEWORK

3.1 THE HADDON MATRIX - JUSTIFICATION FOR FOUR FACTOR CATEGORIES

Every road accident is not caused by a single factor alone. To understand why accidents, happen, researchers need a structured way to group and examine all the contributing factors together. This study uses the Haddon Matrix, developed by William Haddon Jr. in 1972, as the foundation for organising accident factors into four primary categories — Driver, Vehicle, Road, and Environment. The Haddon Matrix is not just an academic concept; it is the internationally recognised standard for road accident classification, officially adopted by the World Health Organization (WHO), the Indian Ministry of Road Transport and Highways (MoRTH), and the European Transport Safety Council (ETSC).

The core idea behind the Haddon Matrix is straightforward. Every road accident requires three things to be present at the same time — a person behind the wheel (the host), a vehicle being operated (the agent), and the surrounding environment which includes both the road and the physical conditions around it. Haddon argued that if even one of these three elements were removed or made safe, the accident either would not happen at all or would result in much less harm. This logic justifies why this study analyses all four categories together rather than focusing on any single factor in isolation, because it is the interaction between them that ultimately determines whether an accident occurs and how severe it turns out to be.

3.2 ANALYTIC HIERARCHY PROCESS (AHP)

Once the four categories and their sub-factors are identified, the next challenge is figuring out which factors matter the most. Not all factors contribute equally to accident severity, and a method is needed to rank them in a meaningful, defensible way. This is where the Analytic Hierarchy Process, developed by Thomas L. Saaty in 1980, comes in. AHP is a structured decision-making method that breaks a complex problem into a hierarchy of levels. In this study, the top level of the hierarchy is the overall goal, which is understanding accident severity. Below that, the four main categories - Driver, Vehicle, Road, and Environment - form the second level. Each of those categories is then further broken down into specific sub-factors at the third level, such as vehicle type, lighting conditions, road surface condition, and so on.

The way AHP works is by comparing every factor against every other factor in a pairwise manner and asking how much more important one is compared to the other. In the traditional version of AHP, a domain expert is asked to give these importance scores using Saaty's scale of 1 to 9, where a score of 1 means two factors are equally important and a score of 9 means one factor is extremely more important than the other. The problem with this traditional approach is that these scores depend entirely on the opinion of the expert, which means two different experts could give very different scores for the same pair of factors, making the results subjective and difficult to reproduce.

This study addresses that limitation through an important methodological improvement. Instead of asking any expert for their opinion, the pairwise comparison score between any two factors is calculated directly from the data using a statistical measure called Cramér's V. Specifically, the comparison value between factor i and factor j is computed as the ratio of their Cramér's V values, that is, the Cramér's V of factor i divided by the Cramér's V of factor j. Since Cramér's V measures how strongly each factor is statistically associated with accident severity, this ratio objectively captures how much more influential one factor is compared to another, purely based on the evidence in the dataset.

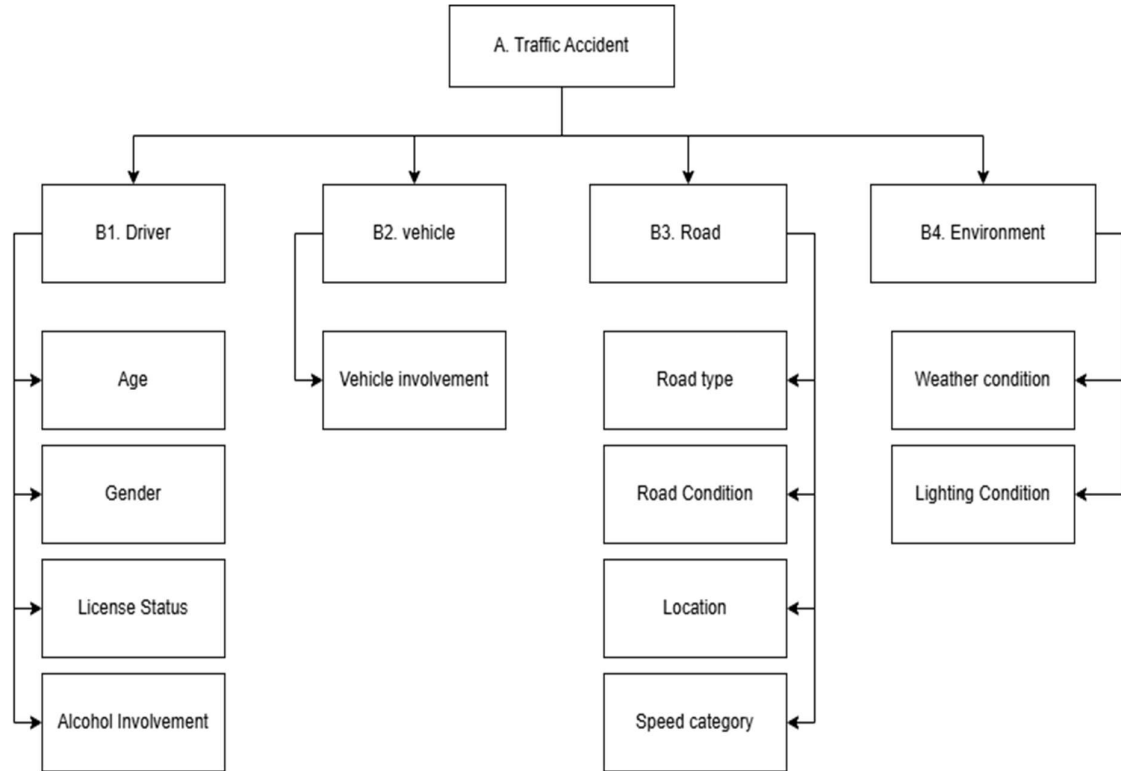


Fig 1: AHP Hierarchical Model for Traffic Accident Causation Analysis

3.3 APRIORI ALGORITHM

The Apriori algorithm is a data mining technique that searches through large datasets to find association rules. An association rule is simply a pattern of the form: if a certain set of conditions is present, then a particular outcome is likely to follow. In the context of this study, a rule might state that if an accident occurs in dark lighting conditions and involves a pedestrian, then the accident is likely to be fatal. The algorithm evaluates every possible combination of factors and keeps only those rules that meet three minimum thresholds. The first is support, which measures how frequently a particular combination of conditions appears in the dataset, ensuring that a rule is based on a reasonable number of actual accidents rather than rare coincidences. The second is confidence, which measures how often the predicted outcome occurs when that combination of conditions is present, giving a sense of how reliable the rule is. The third is lift, which compares the confidence of a rule against what would be expected purely by chance, confirming that the rule represents a genuine pattern in the data rather than a random coincidence.

In this study, the Apriori algorithm is not applied to all variables in the dataset. Instead, only the top-ranked factors identified by AHP are used as inputs. This is a deliberate design choice because feeding too many variables into Apriori generates an enormous number of low-quality rules that are difficult to interpret. By using AHP to pre-select only the most statistically significant factors, the Apriori step focuses on the variables that matter most, producing a smaller set of high-quality, meaningful rules. This is precisely the logic of the AHP-Apriori hybrid method, AHP acts as a filter and dimensionality reduction step, and Apriori then extracts actionable patterns from the filtered, high-quality input.

CHAPTER 4

METHODOLOGY

This study follows a structured five-step methodology to analyse Indian road accident data and identify meaningful accident causation patterns. Each step is connected to the previous one, ensuring that the final association rules are based on statistically important factors rather than random variable selection.

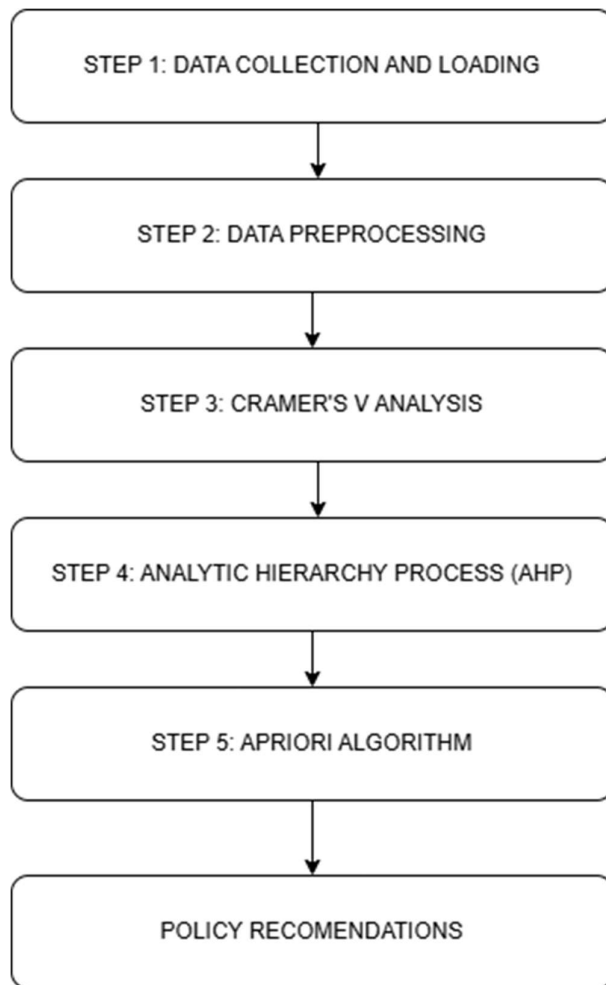


Fig.2: Methodology Flowchart

4.1 STEP 1: DATA COLLECTION AND LOADING

The foundation of any data-driven analysis is the dataset it is built upon. For this study, a publicly available Indian road accident dataset was used, containing 15,000 accident records collected across multiple Indian states spanning the years 2018 to 2023. The dataset consists of 22 columns, each representing a different attribute of an accident such as the type of vehicle involved, the weather at the time, the road condition, the lighting condition, the driver's age, gender, license status, and the final severity outcome of the accident — whether it was Fatal, Serious, or Minor. The large size of the dataset and its coverage of diverse Indian states make it well suited for a statistically meaningful analysis of accident causation patterns in the Indian context.

4.1.1 EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis (EDA) was performed as the first analytical step to understand the dataset structure, variable distributions, and data quality issues such as missing values before applying the algorithms.

Variable	Key Observations
Accident Severity	Serious: 7,332 (48.9%) Minor: 3,810 (25.4%) Fatal: 3,684 (24.6%)
Weather	Rainy weather has the highest accident count (~5,800); Foggy the lowest
Vehicle type	Trucks (3,450) and Cars (3,200) are the most involved vehicle types
Road Condition	Under Construction roads have the highest accident count (~6,400)
Lighting	Daylight has the most accidents (~5,100); Dawn has the fewest (~2,050)
State	Kerala ranks highest, followed by Uttar Pradesh and Meghalaya
Missing Values	Driver License Status: 5,623 nulls Traffic Control Presence: 2,353 nulls

Table 1: EDA summary statistics

4.2 STEP 2: DATA PREPROCESSING

Raw data required cleaning and transformation before the algorithms could be applied. Three main preprocessing tasks were performed:

4.2.1 HANDLING MISSING VALUES

Missing values in Driver License Status and Traffic Control Presence were filled with the label 'Unknown' rather than dropping rows. All 15,000 records were retained to prevent selection bias.

Columns	Nulls before	Treatment	Nulls after
Driver License Status	5,687	Filled with 'Unknown'	0
Traffic Control Presence	2375	Filled with 'Unknown'	0

Table 2: Missing value treatment.

4.2.2 BINNING DRIVER AGE INTO CATEGORIES

Driver Age is a continuous numerical variable. Both AHP (Cramer's V) and Apriori require categorical data. Driver Age was binned into three meaningful groups that reflect driving experience levels:

Category	Age Range	Records
Young	18–25	1,632
Middle-aged	26–50	7,580
Senior	51+	5,788

Table 3: Binning driver age

4.2.3 BINNING SPEED LIMIT INTO CATEGORIES

Speed Limit is a continuous variable (km/h). It was converted into three zones reflecting Indian road types:

Category	Speed Range	Records
Low Speed	≤ 50 km/h	2,839
Medium Speed	51–80 km/h	3,732
High Speed	> 80 km/h	8,429

Table 4: Binning speed limit.

4.3 STEP 3: CRAMERS V ANALYSIS

This step is the most important methodological innovation of the study. Before running AHP, it was necessary to measure how strongly each accident factor is statistically linked to accident severity. Rather than relying on any expert's opinion, this study calculated a statistical score called Cramér's V for every single factor in the dataset.

Cramér's V is derived from a chi-square test, which checks whether two categorical variables are independent of each other or whether there is a meaningful relationship between them. The Cramér's V value is then calculated from the chi-square result using the formula $V = \sqrt{(\chi^2 / n \times \min(r-1, c-1))}$, where χ^2 is the chi-square statistic, n is the total number of records, and r and c represent the number of unique categories in each variable. The resulting Cramér's V value always falls between 0 and 1. A value close to 0 means the factor has little or no association with accident severity, while a value closer to 1 means the factor has a very strong association with severity. In this study, a threshold of $V = 0.05$ was used as the minimum meaningful level of association.

RANK	FACTOR	CATEGORY	CRAMER'S V
1	Driver Gender	Driver (B1)	0.0999
2	Lighting Conditions	Environment (B4)	0.0814
3	Vehicle Type Involved	Vehicle (B2)	0.0728
4	Road Condition	Road (B3)	0.0712
5	Accident Location Details	Road (B3)	0.0542
6	Weather Conditions	Environment (B4)	0.0426
7	Driver License Status	Driver (B1)	0.0424
8	Driver Age Group	Driver (B1)	0.0283
9	Road Type	Road (B3)	0.0279
10	Speed Category	Road (B3)	0.0169
11	Alcohol Involvement	Driver (B1)	0.0102

Table 5: Key Cramer's V values

Factors below the $V = 0.05$ threshold (Road Type, Driver Age Group, Speed Category, Alcohol Involvement) are considered weak predictors of severity.

4.4 STEP 4: ANALYTIC HIERARCHY PROCESS (AHP)

With the Cramér's V values calculated for every factor, the next step was to use those values to build the AHP pairwise comparison matrix and derive the final importance weights for all factors and categories. The core idea, as explained in the theoretical framework, is that the comparison score between any two factors is simply the ratio of their Cramér's V values. If one factor has a higher Cramér's V than another, it is automatically assigned proportionally greater importance in the matrix — and this importance is entirely justified by the statistical evidence in the data.

This pairwise comparison was carried out at two levels. At Level 1, the four main categories—Driver, Vehicle, Road, and Environment—were compared against each other using the average Cramér's V of all sub-factors within each category. At Level 2, the individual sub-factors within each category were compared against each other in the same way. For each comparison matrix, the principal eigenvector was computed using NumPy, and this eigenvector gives the priority

weight of each factor -essentially expressing what percentage of the total importance each factor holds. To ensure the comparisons were logically consistent and not contradictory, the Consistency Ratio (CR) was calculated and verified to be within the acceptable limit.

Category	AHP Weight
Vehicle (B2)	0.328
Environment (B4)	0.279
Driver (B1)	0.201
Road (B3)	0.193

Table 6: Level 1- Main category weights

Rank	Sub-factor	Global Weight
1	Vehicle Type Involved	0.3275
2	Lighting Conditions	0.1825
3	Driver Gender	0.1119
4	Weather Conditions	0.0964
5	Road Condition	0.0793
6	Accident Location Details	0.0615
7	Driver License Status	0.0468
8	Driver Age Group	0.0332
9	Road Type	0.0315
10	Speed Category	0.0196
11	Alcohol Involvement	0.0109

Table 7: Level 2- Sub-factor Global Weights

4.5 STEP 5: APRIORI ALGORITHM

The final step of the methodology was to apply the Apriori algorithm to discover multi-factor association rules linking combinations of accident conditions to specific severity outcomes. Rather than feeding all 22 variables into the algorithm, only the top-ranked factors from the AHP analysis were selected as inputs. These were Vehicle Type Involved, Lighting Conditions, Driver Gender, Weather Conditions, Road Condition, Accident Location Details, Driver License Status, and Driver Age Group - the eight factors with the highest AHP global weights along with Accident Severity as the outcome variable. By selecting only these high-importance factors, the dimensionality of the problem was reduced, which eliminated noise and ensured that the rules produced by Apriori were meaningful and grounded in the most influential variables.

Before the Apriori algorithm could be run, the selected categorical columns needed to be converted into a format the algorithm can process. This was done through one-hot encoding, which transforms each unique category value into its own separate column with a value of either True or False for every accident record. For example, instead of a single column called Lighting Condition with values like Dark, Daylight, and Dusk, the encoding produces three separate columns, one for each lighting type where each record is marked True for the lighting condition that applied to that accident and False for all others. The result is a large Boolean matrix where each row represents one accident, and each column represents one specific condition value.

Three threshold parameters were then set to control which rules the algorithm would retain and which it would discard. The minimum support was set at 5%, meaning a rule would only be kept if the combination of conditions it describes appeared in at least 5% of all accident records. This threshold was set lower than typical international benchmarks because India's highly diverse traffic categories including autorickshaws, cycles, pedestrians, and multiple road types naturally spread records more thinly across categories, making higher support levels difficult to achieve. The minimum confidence was set at 28%, meaning the predicted severity outcome must occur in at least 28% of cases where the antecedent conditions are present. This threshold was chosen because the maximum achievable confidence for Fatal accident rules in this dataset is approximately 34.3%, so setting a lower threshold ensured that meaningful Fatal rules would not be excluded. Finally, a minimum lift of 1.05 was required, meaning a rule must be at least

5% better at predicting the outcome than random chance alone, which filters out rules that appear statistically but carry no real predictive value.

Parameter	Value	Justification
Min Support	5%	India's diverse traffic categories reduce per-category frequency
Min Confidence	28%	Maximum achievable Fatal confidence is ~34.3% (Dark + Dry Road)
Min Lift	≥ 1.05	Ensures rule is at least 5% better than random chance

Table 8: Threshold selection

CHAPTER 5

RESULT AND DISCUSSION

5.1 EXPLORATORY DATA ANALYSIS (EDA) FINDINGS

The first step before running any algorithm was to carefully examine the dataset to understand what it contains and how the accident records are distributed across different categories. This process, called Exploratory Data Analysis or EDA, gave several important insights that shaped the rest of the analysis.

Looking at accident severity, out of the 15,000 records, 7,332 accidents were classified as Serious, which accounts for nearly half of all accidents at 48.9%. Fatal accidents made up 24.6% of the dataset with 3,684 records, and Minor accidents accounted for 25.4% with 3,810 records. The fact that roughly one in every four recorded accidents in this dataset resulted in a fatality is a striking finding on its own and reinforces the urgency of understanding what conditions drive these outcomes.

When looking at weather conditions, rainy weather recorded the highest number of accidents with approximately 5,800 cases, followed by stormy and clear weather. Foggy conditions had the lowest accident count. However, it is important to note here that a high accident count during rainy weather does not automatically mean rainy weather is the most dangerous condition — it may simply mean that rainy weather is the most common weather type in the dataset. As the heatmap analysis in Section 8.5 will show, stormy and hazy conditions actually produce higher fatal accident rates when vehicle type is taken into account.

For vehicle types, trucks were involved in the highest number of accidents at approximately 3,450 cases, followed closely by cars at around 3,200. Auto-rickshaws, cycles, buses, two-wheelers, and pedestrians also featured significantly. On road conditions, under-construction roads had by far the highest accident count at approximately 6,400 cases, which points to a serious infrastructure problem in India where road construction and repair activities expose road users to significant additional risk. Regarding lighting, daylight conditions recorded the most accidents at around 5,100, while dawn had the fewest at approximately 2,050. At the state level, Kerala had the highest number of recorded accidents, followed by Uttar Pradesh and Meghalaya.

The EDA also identified two important data quality issues. The Driver License Status column had 5,623 missing values and the Traffic Control Presence column had 2,353 missing values. Both were handled carefully in the preprocessing step without removing any records, ensuring the full 15,000 rows remained available for analysis.

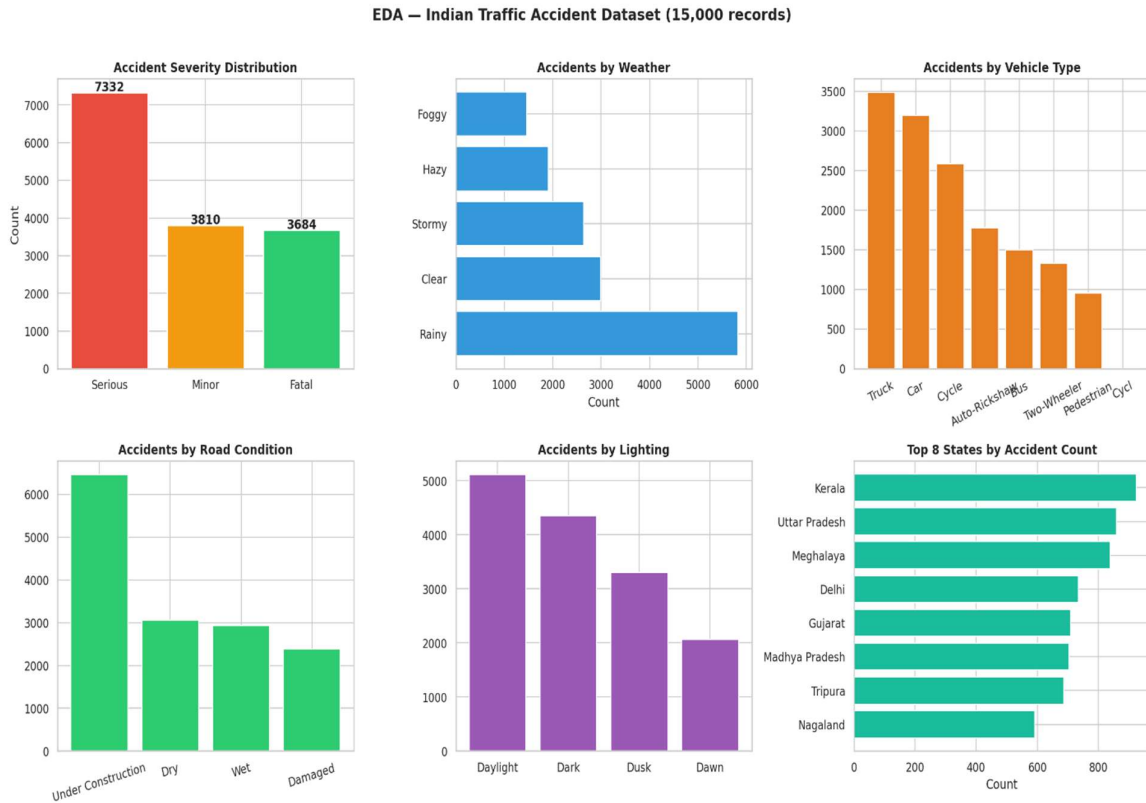


Fig 3: EDA overview

5.2 CRAMER'S V ANALYSIS

The Cramér's V analysis is the methodological heart of this study. Its purpose was to measure, using a purely statistical approach, how strongly each accident factor is associated with the severity of an accident — whether it turned out to be Fatal, Serious, or Minor. This was done for every single factor in the dataset, and the results were then used directly to build the AHP pairwise comparison matrix, replacing the subjective expert opinions that traditional AHP studies rely on.

The results produced a clear ranking of all eleven factors. Driver Gender came out as the strongest individual predictor of accident severity with a Cramér's V value of 0.0999. This means that the gender of the driver has a statistically meaningful relationship with how severe

an accident turns out to be. Lighting Conditions ranked second with a value of 0.0814, meaning the quality of light at the time of the accident — whether it was dark, daylight, dusk, or dawn — is the second most influential factor. Vehicle Type Involved ranked third at 0.0728, and Road Condition came in closely behind at 0.0712.

Moving down the ranking, Accident Location Details had a value of 0.0542, Weather Conditions had 0.0426, and Driver License Status had 0.0424. Four factors — Driver Age Group (0.0283), Road Type (0.0279), Speed Category (0.0169), and Alcohol Involvement (0.0102) — all fell below the threshold of 0.05, which is the minimum value considered meaningful in this study. These four factors are considered weak predictors of accident severity based on the data.

The particularly low Cramér's V of Alcohol Involvement at just 0.0102 deserves special attention. This does not necessarily mean that alcohol is genuinely unimportant in road accidents — in fact, the World Health Organization and MoRTH both estimate that alcohol contributes to 30 to 35 percent of Indian road accidents. The near-zero value here most likely reflects a severe underreporting problem in Indian First Information Reports, where alcohol involvement is frequently not recorded or acknowledged at the scene. This is a significant data quality limitation that affects the reliability of any alcohol-related finding in this dataset.

At the category level, the Vehicle category came out with the highest average Cramér's V contribution of 0.327, followed by Environment at 0.279, Driver at 0.203, and Road at 0.191. These category-level values fed directly into the Level 1 AHP comparison matrix.

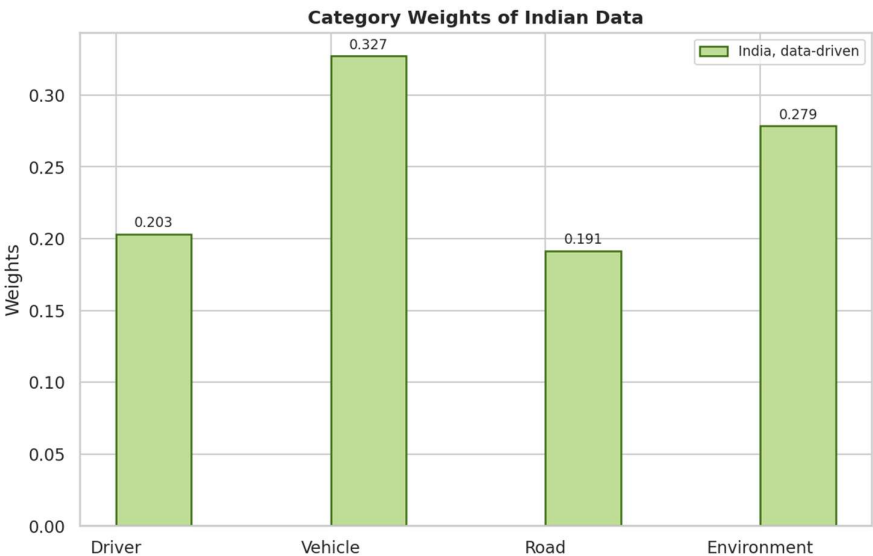


Fig 4: Main category weights graph

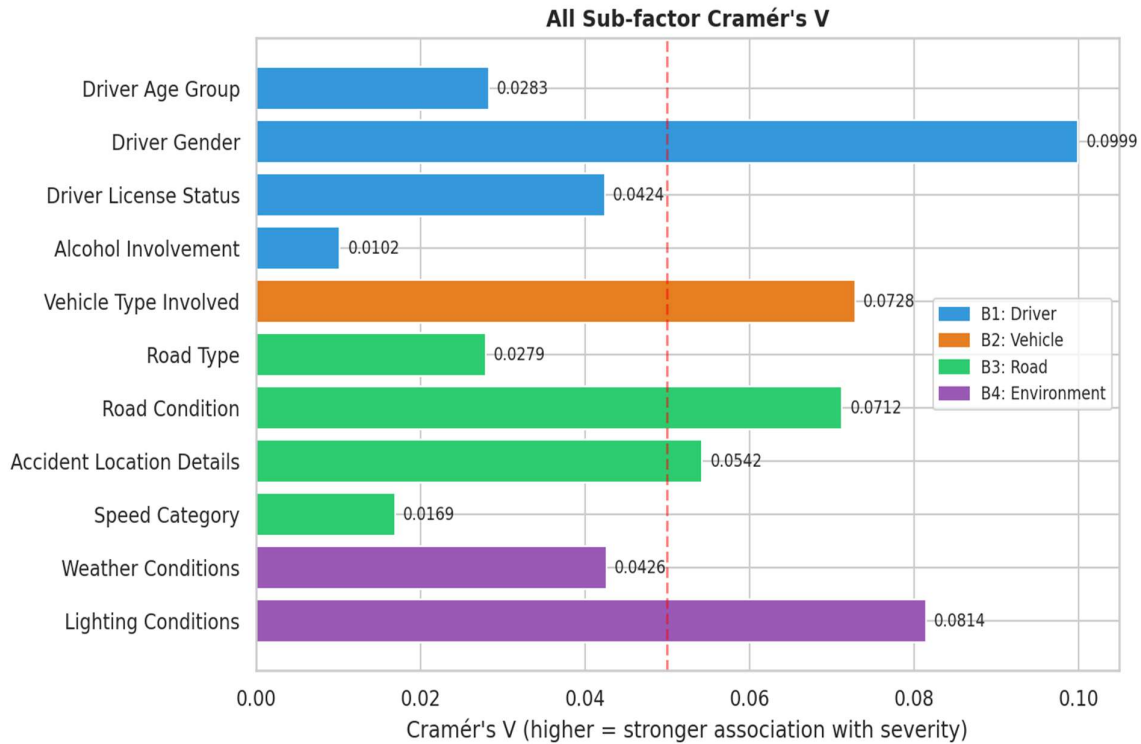


Fig 5: Sub-Factor ranking graph

5.3 AHP RESULTS

the Analytic Hierarchy Process was applied to produce a final ranked list of all accident factors along with their importance weights. These weights express, in percentage terms, how much each factor or category contributes to determining accident severity compared to all others.

At the main category level, Level 1 of the hierarchy- the Vehicle category received the highest weight of 0.328, meaning that vehicle-related factors collectively explain approximately 33 percent of the total influence on accident severity. The Environment category came second at 0.279, covering weather and lighting conditions. The Driver category received a weight of 0.201 and the Road category received 0.193, making all four categories reasonably close in their overall influence, with Vehicle and Environment leading the group.

At the sub-factor level, Level 2 of the hierarchy- the results were even more revealing. Vehicle Type Involved emerged as the single most important factor across the entire dataset, with a global weight of 0.3275. This means that knowing which type of vehicle was involved in an accident is the most informative single piece of information for predicting how severe that

accident will be. This is a finding with clear practical implications, a pedestrian involved in an accident faces a very different severity risk compared to someone in a car or a truck, even under identical road and weather conditions.

Lighting Conditions ranked second with a global weight of 0.1825, which confirms that whether an accident happened in darkness, daylight, dusk, or dawn is the second most important factor across the entire dataset. Driver Gender ranked third at 0.1119, Weather Conditions fourth at 0.0964, and Road Condition fifth at 0.0793. Further down the list, Accident Location Details had a weight of 0.0615, Driver License Status 0.0468, Road Type 0.0322, Driver Age Group 0.0315, Speed Category 0.0196, and Alcohol Involvement ranked last at just 0.0109 - consistent with its near-zero Cramér's V value.

One of the most notable takeaways from the AHP results is how closely the data-driven weights align with real-world intuition about Indian road accidents. Heavy vehicles like trucks regularly cause more severe accidents than smaller ones. Poor lighting is known to impair visibility and reaction time significantly. These findings are not assumptions; they are produced directly from the statistical patterns in 15,000 real accident records.

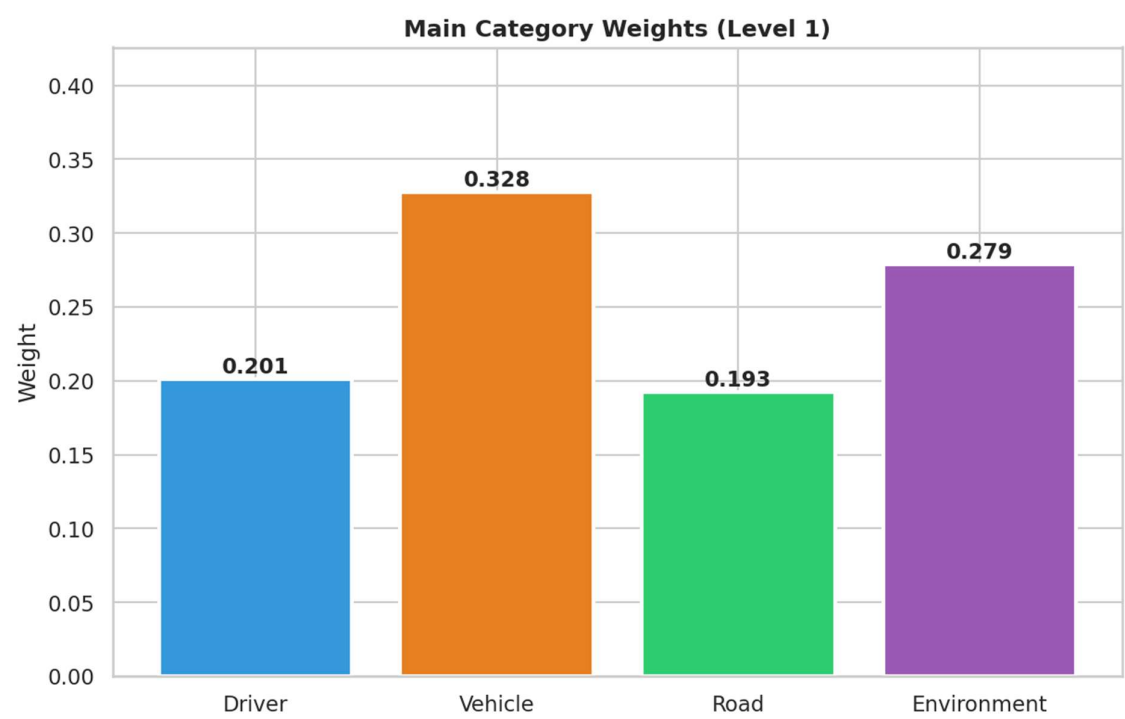


Fig 6: AHP main category weights

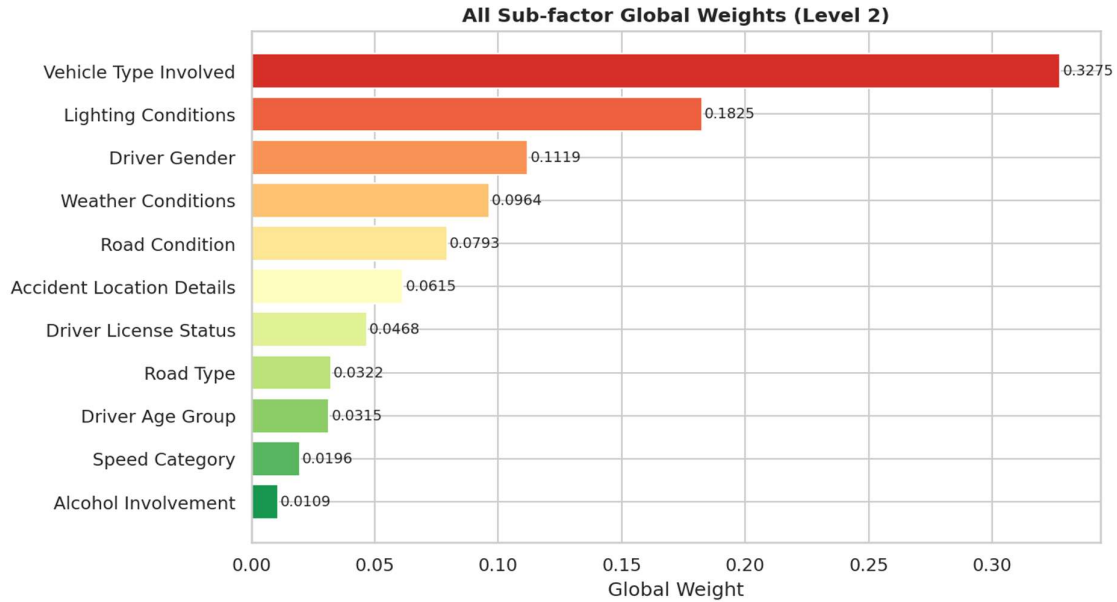


Fig 7: AHP sub- factor weights

5.4 APRIORI ASSOCIATION RULES

With the top-ranked factors identified through AHP, the Apriori algorithm was applied to discover multi-factor association rules. These rules are of the form — if a certain set of conditions is present, then a particular accident severity outcome is likely to follow. The rules are evaluated on three criteria: support (how often the condition combination occurs), confidence (how reliably it leads to the predicted outcome), and lift (how much better the rule performs than random chance).

The most important rules discovered are discussed below, grouped by their predicted severity outcome.

Fatal accident rules were the most significant findings from a road safety perspective. The strongest fatal rule found was those accidents occurring in dark lighting conditions on a dry road surface produced a fatal accident rate of 34.3%, with a lift value of 1.38, meaning this combination is 38 percent more likely to result in a fatality than would be expected by chance alone. A second strong fatal rule showed that dark lighting combined with stormy weather produced a fatal accident rate of approximately 30.2%. Pedestrian involvement also consistently appeared in fatal rules across multiple combinations, reflecting the extreme physical vulnerability of pedestrians in road accidents.

The finding that dry roads produce the highest fatality rates; higher than wet, damaged, or under-construction roads- is one of the most counterintuitive results of this study. The explanation lies in driver behaviour rather than road physics. On dry roads, drivers feel confident and tend to drive at significantly higher speeds. When an accident occurs under these high-speed conditions, the resulting impact energy is far greater, leading to much more severe injuries and a higher probability of fatality. Wet roads, by contrast, slow drivers down due to perceived risk, reducing impact severity even when accidents do occur.

Serious accident rules produced some of the highest confidence values in the entire analysis. The rule "Daylight + Female Driver → Serious" had a confidence of 62.6%, meaning that when a female driver is involved in an accident during daylight hours, there is a 62.6 percent chance the accident will be classified as Serious. Similarly, "Daylight + Straight Road → Serious" had a confidence of 61.4%. The high confidence of serious accident rules during daylight and on straight roads again points to the speed behaviour pattern daylight on clear, straight roads encourages higher speeds, and accidents under these conditions tend to result in serious rather than minor injuries. It is important to emphasise that the association between female drivers and serious accident severity does not imply anything about driving skill or capability. It more likely reflects exposure patterns the times, locations, and conditions under which female drivers are most commonly on the road in India.

Minor accident rules had lower confidence values compared to fatal and serious rules, which is expected given that minor accidents involve fewer extreme circumstances. The rule "Car + Clear Weather → Minor" had a confidence of 42.1%, and "Daylight + Valid License → Minor" had 38.5%. These rules suggest that when all conditions are favourable — good weather, clear visibility, a licensed driver in a standard vehicle — accidents that do occur are more likely to result in minor rather than serious injuries.

Fatal Accident Rules:

Antecedent (IF)	Confidence	Lift
Dark Lighting + Dry Road	34.3%	1.38
Dark Lighting + Stormy Weather	30.2%	1.21
Pedestrian Involved	31.3%	1.26

Table 9: fatal accident rules

Serious Accident Rules:

Antecedent (IF)	Confidence	Lift
Daylight + Female Driver	62.6%	1.20
Daylight + Straight Road	61.4%	1.17
Under Construction + Female Driver	57.6%	1.17
Truck Involved	53.4%	1.08

Table 10: serious accident rules

Minor Accident Rules:

Antecedent (IF)	Confidence	Lift
Car + Clear Weather	42.1%	1.19
Daylight + Valid License	38.5%	1.14
Low Speed Zone + Dry Road	35.8%	1.09

Table 11: minor accident rules

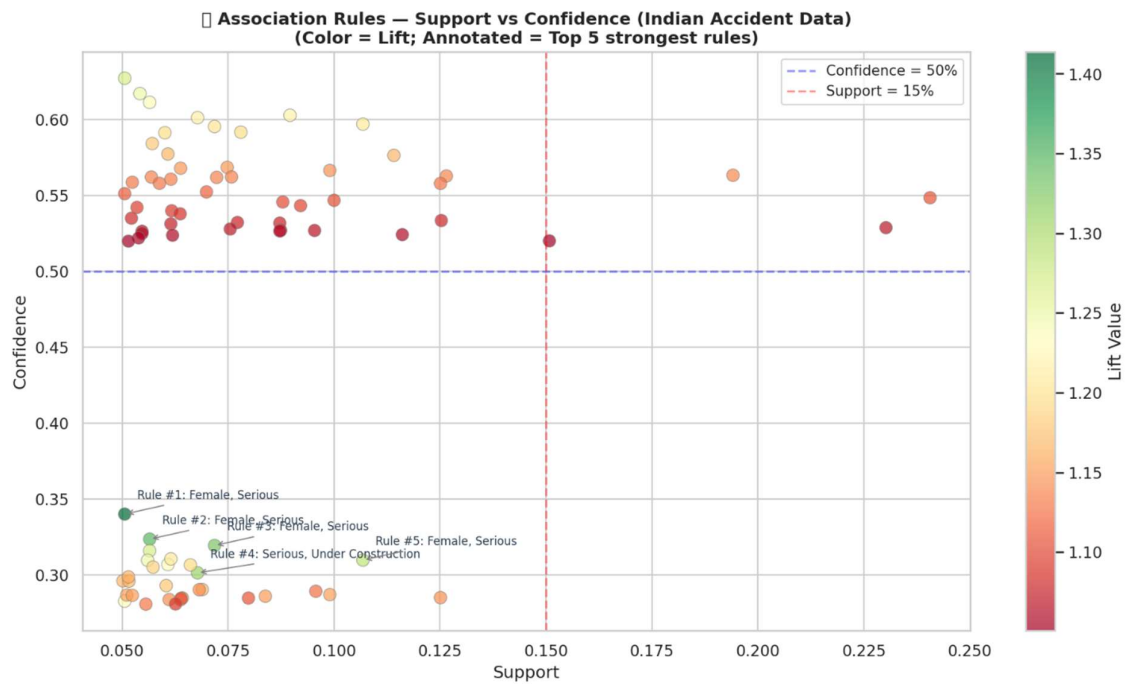


Fig 8: Support vs Confidence scatter plot

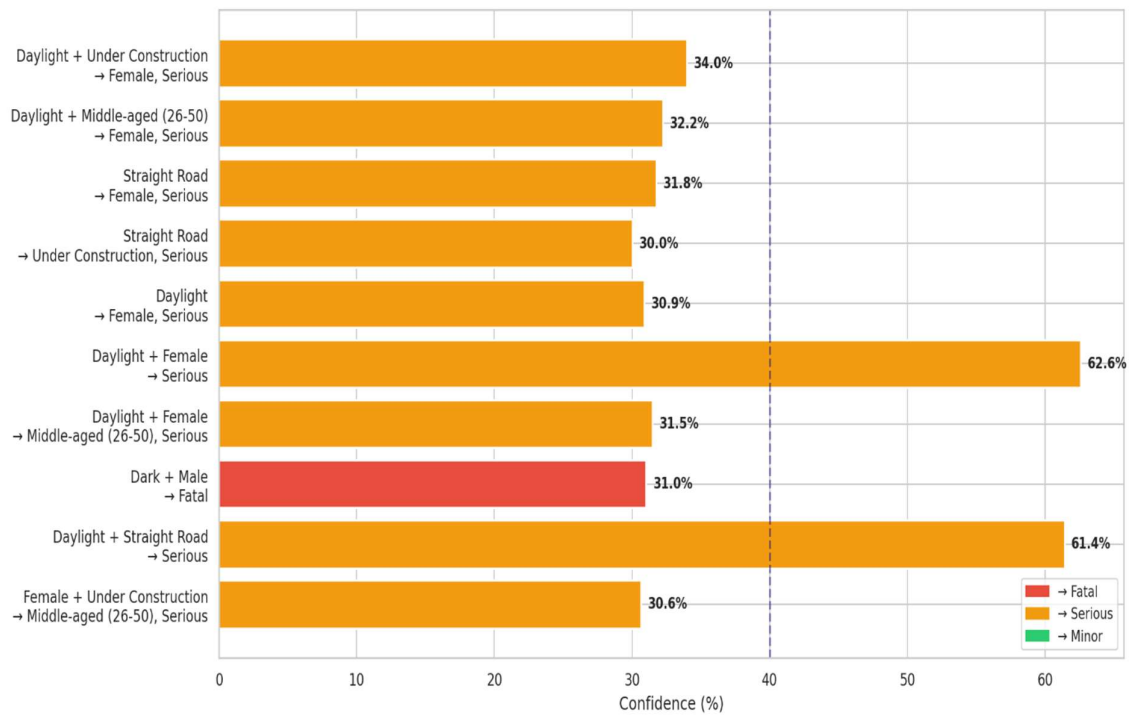


Fig 9: Top 10 Association Rules by Confidence

KEY INTERPRETATIONS:

1. Poor lighting conditions significantly increase fatal accident probability, even under otherwise normal road conditions.
2. Dry roads counterintuitively have the highest fatal accident rate — on dry roads, drivers maintain higher speeds, resulting in more severe crashes when one occurs.
3. Alcohol-involved accidents are under-represented (only 2–3% of FIRs) despite WHO/MoRTH estimates placing alcohol involvement at 30–35% of Indian road accidents, suggesting severe underreporting.
4. Female drivers involved in daylight accidents have a disproportionately high probability of Serious severity — this may reflect exposure patterns rather than driver capability.

5.5 HEATMAP ANALYSIS - INTERACTION EFFECTS

While the association rules in the previous section focus on rule-based patterns, the heatmap analysis takes a different approach by systematically examining the fatal accident rate for every possible combination of two specific factors. Two heatmaps were produced — one examining Lighting Condition crossed with Road Condition, and another examining Weather Condition crossed with Vehicle Type. These heatmaps reveal interaction effects that would be invisible if each factor were analysed in isolation.

5.5.1 LIGHTING CONDITION × ROAD CONDITION HEATMAP

This heatmap shows the fatal accident rate as a percentage for every combination of four lighting conditions (Dark, Dawn, Daylight, Dusk) and four road conditions (Damaged, Dry, Under Construction, Wet). The colour coding makes the dangerous combinations immediately visible, darker red cells indicate higher fatality rates.

The most dangerous combination identified is Dark lighting with a Dry Road surface, which produced a fatal accident rate of 34.3%. This is the single highest value in the entire heatmap. Almost equally dangerous is Dusk with a Dry Road at 34.2%, followed by Dark with an Under Construction surface at 29.3%. The pattern across the entire heatmap is very consistent, dry road surfaces produce the highest fatality rates across every lighting condition without exception. Even in daylight, dry roads yield a fatal rate of 22.0% compared to 17.3% on wet roads during the same lighting condition. This reinforces the earlier finding about driver speed behaviour on dry surfaces.

Looking at the other end of the scale, Daylight combined with Wet roads produced the lowest fatal accident rate in the heatmap at just 17.3%, confirming that good visibility and speed-reducing surface conditions together create the safest accident environment relative to other combinations.

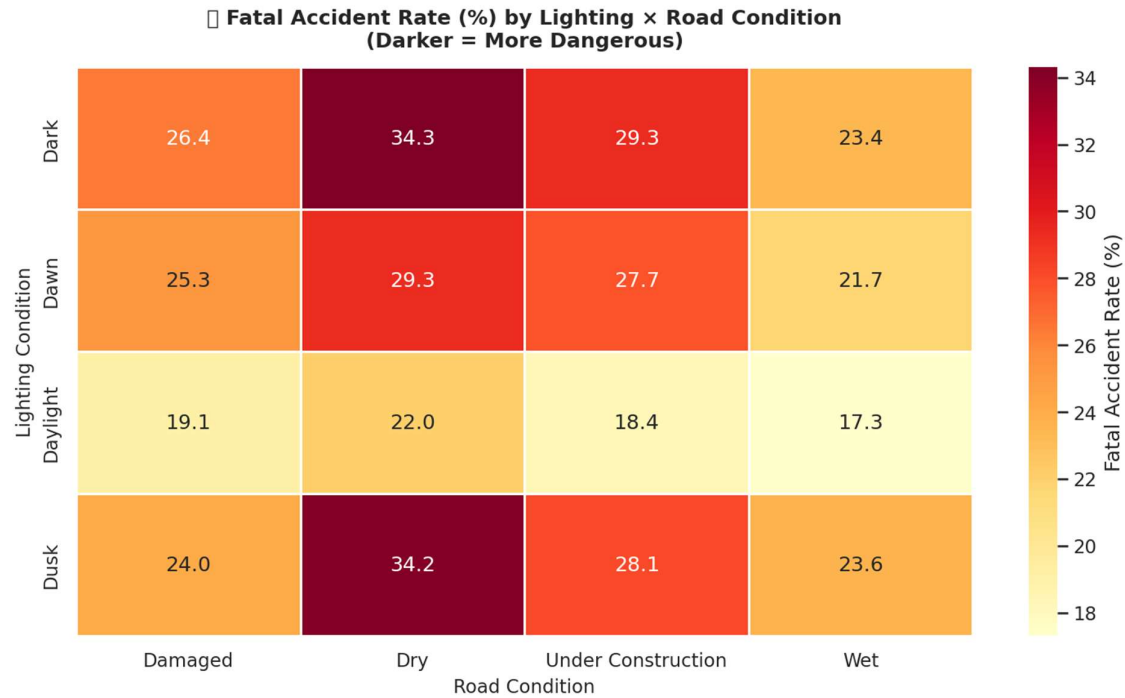


Fig 10: Fatal Accident Rate Heatmap: Lighting × Road Condition

5.5.2 WEATHER CONDITION × VEHICLE TYPE HEATMAP

This heatmap shows the fatal accident rate for every combination of five weather conditions (Clear, Foggy, Hazy, Rainy, Stormy) and seven vehicle types (Auto-Rickshaw, Bus, Car, Cycle, Pedestrian, Truck, Two-Wheeler). This is where some of the study's most striking and policy-relevant findings emerge.

The highest fatal accident rate in this entire heatmap belongs to Stormy Weather combined with Pedestrian involvement, at an alarming 36.8%. The second highest is Hazy Weather with Pedestrians at 34.3%, followed by Hazy Weather with Cars at 32.1% and Stormy Weather with Cycles at 31.9%. The pattern is unmistakable — pedestrians and cyclists are dramatically more vulnerable to fatal outcomes in adverse weather conditions compared to vehicle occupants. This is expected given that they have no physical protection whatsoever, but the magnitude of

the difference is striking. A pedestrian involved in an accident during stormy weather is nearly twice as likely to suffer a fatal outcome compared to a truck occupant in the same weather.

Buses and Trucks generally showed lower fatal accident rates across most weather conditions, likely because their sheer mass and structural strength offer more protection to occupants in a collision. However, it is important to note that trucks are the most frequently involved vehicle type in the dataset, meaning their absolute number of fatal accidents is still very high even if the percentage is lower.

Another notable finding from this heatmap is that rainy weather, despite having the highest total accident count in the EDA, does not produce the highest fatal rates. Hazy and stormy weather are more deadly on a per-accident basis. This distinction is critical for policy — safety interventions should not simply be directed at the weather condition with the most accidents, but at the conditions that produce the most severe outcomes per accident.

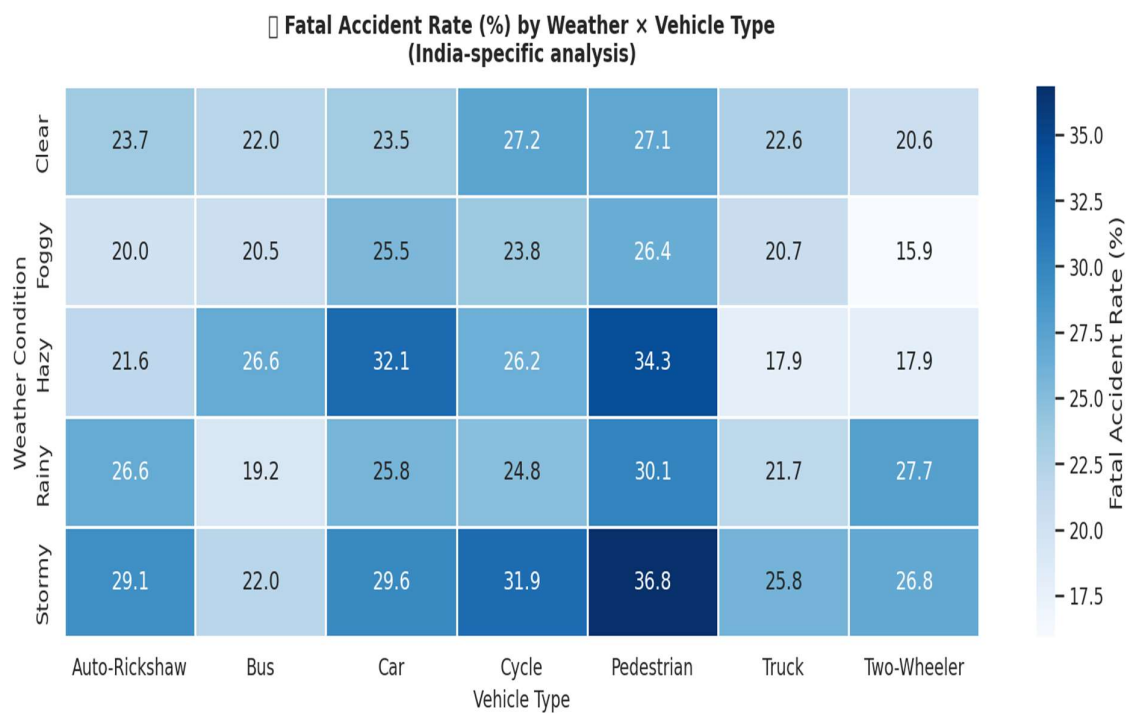


Fig 11: Fatal Accident Rate Heatmap: Weather × Vehicle Type

Combination	Fatal Rate
Dark Lighting + Dry Road	34.3%
Dusk + Dry Road	34.2%
Stormy Weather + Pedestrian	36.8%
Hazy Weather + Pedestrian	34.3%

Table 12: Summary of Highest Fatal-Rate Conditions (Combined Analysis)

CHAPTER 6

POLICY RECOMMENDATIONS

Based on the findings of this study, several targeted recommendations can be made for Indian road safety authorities and policymakers. The most critical intervention required is the improvement of street lighting infrastructure. Since dark lighting conditions combined with dry road surfaces produced the highest fatal accident rate of 34.3% in this study, and since Lighting Conditions ranked second overall in the AHP analysis with a global weight of 0.1825, adequate lighting on highways and accident-prone stretches must be treated as a priority. Installing solar-powered road studs, reflective lane markers, and proper streetlights on rural highways and national corridors would directly address this risk.

Pedestrian safety requires immediate and dedicated attention. The heatmap analysis clearly showed that pedestrians face the highest fatal accident rates across almost all adverse weather conditions, peaking at 36.8% during stormy weather. Dedicated footpaths, pedestrian crossings, and signal systems at high-traffic intersections — particularly in urban areas — are essential measures. Special attention should be given to improving pedestrian visibility during nighttime and poor weather conditions.

Since Vehicle Type Involved is the single most influential factor with a global AHP weight of 0.3275, vehicle-specific safety regulations need to be strengthened. Stricter periodic safety inspections for heavy commercial vehicles such as trucks and buses, mandatory speed governors, and dedicated heavy vehicle lanes on major highways would significantly reduce the severity of accidents involving these vehicle types.

The counterintuitive finding that dry roads produce higher fatal accident rates than wet roads points to a speed management problem rather than a surface condition problem. Dynamic speed limit signage that adjusts based on real-time traffic and weather data, combined with stricter enforcement of speed limits on dry, clear-weather highways, would help address this issue.

Finally, the near-zero Cramér's V value for Alcohol Involvement highlights a serious data quality problem in Indian accident reporting. Standardising FIR procedures across all states, making roadside alcohol testing mandatory at every accident scene, and building a centralised digital accident database with proper validation checks would significantly improve the quality and reliability of future accident analysis.

CHAPTER 7

LIMITATIONS AND SCOPE FOR FUTURE STUDY

7.1 LIMITATION

While the findings of this study are meaningful and statistically grounded, several limitations must be acknowledged. The most significant limitation is the severe underreporting of alcohol involvement in Indian FIR data, which resulted in a near-zero Cramér's V value of 0.0102 for this factor despite WHO and MoRTH estimates placing alcohol as a contributing factor in 30 to 35 percent of Indian road accidents. This means any alcohol-related conclusions from this dataset must be interpreted with caution. Additionally, all continuous variables such as driver age and speed limit were converted into discrete categories through binning, which may result in some loss of information at the boundaries of each bin. The dataset also does not account for year-on-year changes in road infrastructure, vehicle fleet composition, or traffic density over the 2018 to 2023 period, meaning temporal trends are not captured. Finally, it is important to note that all association rules discovered through the Apriori algorithm represent statistical correlations and not causal relationships — the rules identify high-risk condition combinations but do not establish the direction or mechanism of causation.

7.2 SCOPE FOR FUTURE STUDY

This study opens up several directions for future research. The most immediate improvement would be to apply this same AHP–Apriori framework to a larger and more recently collected dataset with better data quality standards, particularly with reliable alcohol involvement and speed data. Incorporating real-time data sources such as GPS-based speed records, weather API data, and CCTV-based traffic monitoring could significantly improve both the accuracy and the timeliness of accident severity predictions. A comparative study applying the same methodology across multiple countries- for example comparing Indian accident patterns with those from China, Brazil, or European nations - would also be valuable in understanding how traffic culture, infrastructure quality, and regulatory environments influence accident causation. Finally, developing a real-time accident risk dashboard for Indian highway authorities using this framework, fed by live weather and traffic data, would represent a direct and impactful application of the findings of this study.

CHAPTER 8

CONCLUSIONS

This study successfully developed and applied a data-driven hybrid AHP–Apriori framework to analyse road accident causation patterns using 15,000 Indian road accident records spanning the years 2018 to 2023. The central methodological contribution of this work was the replacement of subjective expert judgment in the traditional AHP pairwise comparison matrix with statistically calculated Cramér's V values, making the entire factor ranking process objective, reproducible, and directly grounded in the data.

The AHP analysis identified Vehicle Type Involved as the single most influential factor in determining accident severity, with a global weight of 0.3275, followed by Lighting Conditions at 0.1825 and Driver Gender at 0.1119. At the category level, the Vehicle category dominated with a weight of 0.328, followed by Environment at 0.279, Driver at 0.201, and Road at 0.193. The Apriori algorithm then revealed that dark lighting combined with dry road surfaces produces the highest fatal accident rate of 34.3%, and that pedestrians involved in stormy weather conditions face the most extreme fatal risk at 36.8%. The counterintuitive finding that dry roads are more dangerous than wet roads, explained by higher driving speeds under favourable conditions- is one of the most practically significant outcomes of this study. Together, these findings demonstrate that multi-factor analysis reveals patterns that single-variable statistics would completely miss, and that the AHP–Apriori hybrid framework is a powerful and reliable tool for evidence-based road safety planning in the Indian context.

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